KARTHIKEYAN A S

PhD Research Scholar | Computer Vision | Embedded Systems

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RESEARCH PROFILE

PhD Research Scholar specializing in **Computer Vision, Biomechanics, and Embedded Systems**, with experience in developing **AI-driven multi-view tracking systems, sports analytics platforms, and smart sensing devices**. Strong background in integrating **hardware systems with real-time intelligent analytics**.

WORK EXPERIENCE

PhD RESEARCH SCHOLAR **(Jul-2022 To Till date)**

INDIAN INSTITUTE OF TECHNOLOGY, MADRAS, CHENNAI.

Department of Applied Mechanics & Biomedical Engineering

- Conducting interdisciplinary research in **biomechanics, computer vision, and intelligent sensing systems**
- Developed a **multi-view athlete tracking system (Top-view + Side-view)** for robust identity preservation under occlusion
- Designed and implemented **real-time boxing analytics pipeline**, including:
- **Pose estimation (YOLOv8-Pose)**
- **Multi-object tracking (DeepSORT + custom ID logic)**
- **Punch detection and classification (left/right, hook/straight/uppercut, head/body)**
- Engineered **tactical inference modules** for stance, distance, and engagement analysis in combat sports
- Integrated **semantic segmentation + HSV-based color modeling** for robust athlete identification

TEACHING ASSISTANT – NPTEL **(Jul-2025 To Oct-2025)**

INDIAN INSTITUTE OF TECHNOLOGY, MADRAS, CHENNAI.

Course: ***Statistics for Biomedical Engineers***

- Assisted in **teaching, doubt clarification, and student mentoring**
- Supported **assignment evaluation and course management**
- Contributed to **large-scale online education (NPTEL platform)**

PROJECT ASSOCIATE **(Apr-2021 To Jul-2022)**
INDIAN INSTITUTE OF TECHNOLOGY, MADRAS, CHENNAI.

- Designed and developed **Wi-Fi enabled embedded systems (DOT MCU platform)**
- Built **PCB designs and firmware (Arduino + ESP-based IoT systems)**
- Developed **hardware prototypes with 3D printing (PLA/SLA)**
- Contributed to **research-driven system development and experimentation**

RESEARCH & DEVELOPMENT ENGINEER **(Feb-2021 To Apr-2021)**
PRESEVI INDUSTRIES PVT LTD, CHENNAI.

- Handled complete circuit & PCB design of MCU based display boards.
- Have experience in reverse engineering PCB.
- Designs and performs experimental studies of R&D projects
- Have hands-on experience in power circuit design.

INTERNSHIP EXPERIENCE

EMBEDDED HARDWARE AND PCB DESIGNER INTERN **(Jan-2020 To Mar-2020)**
University Of Potential Excellence, Anna University.

- Have hands-on experience in working on semi-automated antibiogram device
- Prototyping of PCB with single and double layers using the Laser etching method.
- Handled complete circuit & PCB design of antibiogram device.
- Have hands-on experience in SMD Soldering.
- Simultaneous testing and performance checking of the antibiogram device.

ACADEMIC RECORD

Course	Institution	Board/ University	Completion By	Marks (%)
X	Little Flower Hr. Sec. School, Salem-636007.	State	2012	61.8
DECE	Muthayammal Polytechnic College, Namakkal-637408.	DOTE	2015	94.5
B.E (ECE)	PSG College of Technology, Coimbatore-04.	Anna University	2018	6.42
M.E (VLSI DESIGN)	Anna University, College of engineering, Chennai-25.	Anna University	2020	7.64
PhD	Indian Institute of Technology, Madras	IIT Madras	2022*	8.00

AREAS OF INTEREST

- Internet Of Things
- Embedded AI System
- Computer Vision
- Sports Analytics

SKILL SET

Programming	Python, C, Embedded C, verilog, VHDL, MicroC
Computer Vision	YOLO, Pose Estimation, Multi-object Tracking
Platforms	Linux, Windows, Raspbian
Tools	MATLAB, keil IDE v5, Tanner IDE, ALTIUM, eSim, ki-cad, EasyEDA
Technologies	Embedded, VLSI, PCB design, circuit design Xilinx FPGA

RESEARCH PROJECTS

Smart Boxing Analytics System (Granted Patent + Publications)

- Developed an **AI-driven multi-camera analytics framework (Top-view + Side-view)** for comprehensive monitoring of boxing bouts
- Designed a **robust multi-object tracking pipeline** combining **YOLO-based detection, pose estimation (keypoints), and DeepSORT with custom ID consistency logic**
- Implemented **temporal tracking and bout segmentation** for long-duration video analysis under real-world conditions
- Engineered a **fine-grained punch detection and classification module**, including:
    - **Hand identification** (left/right)
    - **Punch type classification** (jab, cross, hook, uppercut)
    - **Target zone detection** (head vs body) using opponent keypoints and beltline estimation
- Developed **tactical analytics modules** to extract high-level insights such as:
    - Fighter stance (orthodox/southpaw)
    - Inter-fighter distance and ring control
    - Engagement patterns and activity levels
- Integrated **semantic segmentation and HSV-based color modeling** for **robust boxer identity assignment**, minimizing ID switches under occlusion and camera viewpoint variations
- Designed a **multi-view fusion strategy** to maintain identity consistency across top-view and side-view streams
- Built a **real-time visualization system** with animated overlays and statistical summaries for performance feedback

Padhugai Smart Insole System (Patent Filed)

- Developed a **smart insole platform for gait and posture analysis** using embedded sensors
- Designed **sensor integration and data acquisition system** for pressure and motion tracking
- Implemented **real-time data processing and wireless transmission (IoT-based)**
- Extracted **biomechanical parameters** such as gait patterns, variability, and asymmetry
- Targeted applications in **sports performance analysis and healthcare monitoring**

IoT-Based Gas Leakage Detection System

- Designed an **IoT-enabled safety system** for real-time gas leakage detection
- Integrated sensors with **microcontroller and wireless communication modules**
- Enabled **remote monitoring and alert system (mobile/PC interface)**

Automated Antibiogram Sample Filling System (96-Well Plate)

- Developed a **microcontroller-based automation system** for medical sample handling
- Reduced **manual intervention, processing time, and contamination risk**
- Designed complete **hardware control system and actuation logic**

Automobile Safety and Security System

- Built a **multi-sensor safety system** using proximity and biometric authentication
- Integrated **fingerprint sensor + microcontroller-based control logic**
- Designed to **prevent unauthorized access and reduce accident risk**

Footstep Power Generation System

- Designed an **energy harvesting system** using vibration-based sensing
- Implemented **microcontroller-based power management (PIC + MPLAB)**
- Demonstrated **conversion of mechanical energy into usable electrical power**

Bank Locker Security System

- Developed a **multi-layer security system** with alarm and notification features
- Integrated **microcontroller, LCD display, and alert mechanisms (SMS-based)**
- Enhanced **security monitoring and intrusion detection**

PUBLICATIONS

- ***HistoTrack++: Vision-Based System for Temporal Bout Segmentation and Multi-Target Tracking*** Journal of Signal Processing Systems, 2026
- ***AI-Based Longitudinal Performance Monitoring of Multiple Boxers*** International Conference, 2024
- ***Towards AI-Enabled Automated Tracking of Multiple Boxers*** arXiv, 2023

ACADEMIC ACHIEVEMENTS

- Presented a paper in State level Technical Symposium on"**4G TECHNOLOGY**" at JAYAM POLYTECHNIC COLLEGE.
- Presented a paper in State level Technical Symposium on "**HUMANOID ROBOT**" at THIAGARAJAR POLYTECHNIC COLLEGE.Attended one day of the workshop on **"ARDUINO UNO"** in KONGU Engineering College.

EXTRA-CURRICULAR ACTIVITIES

- Secured **3rd** place in district level science exhibition.
- Secured **2nd** place in pencil sketching competition in PSG tech.

REFERENCES

- Dr.Babji Srinivasan

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Place: Chennai

Date: (Karthikeyan A S)